

EDS 213: Databases and Data Management

Week 1 Assignment

1. Create an ER Diagram

This assignment creates an ER diagram for the Harry Potter database. The five required tables are student, house, wand, course and enrollment.

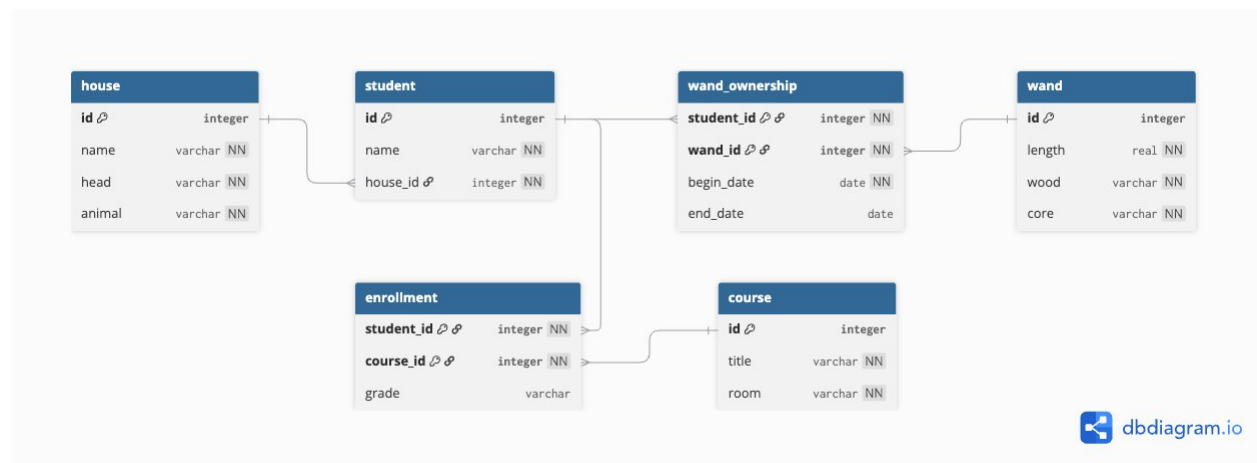
Twist 1: Model a many-to-many relationship between student and wand. The same wand can be used by different students, and one student may own multiple wands. This requires an intermediate table, called wand_ownership, following the same pattern as the enrollment table.

Twist 2: Store the begin date and end date of wand ownership. Because these dates describe the relationship between a student and a wand, they belong in wand_ownership, not in student or wand alone. This follows the same logic as grade belonging in enrollment rather than in student or course.

Solution

ER Diagram

I made a diagram with six tables: the five required tables plus wand_ownership. The diagram shows all six tables with their attributes, primary keys, foreign keys, and cardinality notation. A single bar represents the one side of a relationship and a crow's foot represents the many sides.



Tables and Attributes

Primary keys are marked PK. Foreign keys are marked FK.

PK, FK means the column is both part of the primary key and a foreign key.

house

Stores the four Hogwarts houses. Primary key: `id`.

- `id` (PK)
- `name`
- `head`
- `animal`

student

Stores each student. `house_id` links each student to their house. Primary key: `id`.

- `id` (PK)
- `name`
- `house_id` (FK)

wand

Stores each wand. Ownership is not stored here because the relationship is many-to-many and is handled by `wand_ownership`. Primary key: `id`.

- `id` (PK)
- `length`
- `wood`
- `core`

wand_ownership

The intermediate table that connects `student` and `wand` for the many-to-many relationship. It also stores the date range of ownership. Primary key: the combination of `student_id` and `wand_id`.

- `student_id` (PK, FK)
- `wand_id` (PK, FK)
- `begin_date`
- `end_date`

course

Stores each Hogwarts course. Primary key: `id`.

- `id` (PK)
- `title`
- `room`

enrollment

The intermediate table connecting `student` and `course`. `grade` belongs here because it describes the student course relationship, not either entity alone. Primary key: the combination of `student_id` and `course_id`.

- `student_id` (PK, FK)

- `course_id` (PK, FK)
- `grade`

Design Decisions

Twist 1: Many-to-many between student and wand

The same wand can be owned by different students at different times, and one student can own multiple wands. Because both directions are many, a single foreign key column in either table cannot represent this. The `wand_ownership` table holds `student_id` and `wand_id` as foreign keys, creating the many-to-many link. The crow's foot appears on both sides entering `wand_ownership`.

Twist 2: Where do `begin_date` and `end_date` belong?

`begin_date` and `end_date` describe when a specific student owned a specific wand. They describe the relationship between a student and a wand, not either entity on its own.

`wand_ownership` represents that relationship, so the dates belong there. This is the same reason `grade` belongs in `enrollment` rather than in `student` or `course`.

Primary key of `wand_ownership`

The primary key is the combination of `student_id` and `wand_id`. This follows the same pattern as `enrollment`, where the primary key is the combination of `student_id` and `course_id`. No separate `id` column is added.

One ambiguity here is whether a student could own the same wand more than once at different times. For example, borrow it, return it, and borrow it again. If that is possible, (`student_id`, `wand_id`) would not be unique and the primary key would break. One way to handle this would be to add `begin_date` to the primary key, making it (`student_id`, `wand_id`, `begin_date`). Since the assignment does not specify this case, I kept the simpler two column key, but this is a real ambiguity that would need to be resolved with more information about how wand ownership works.